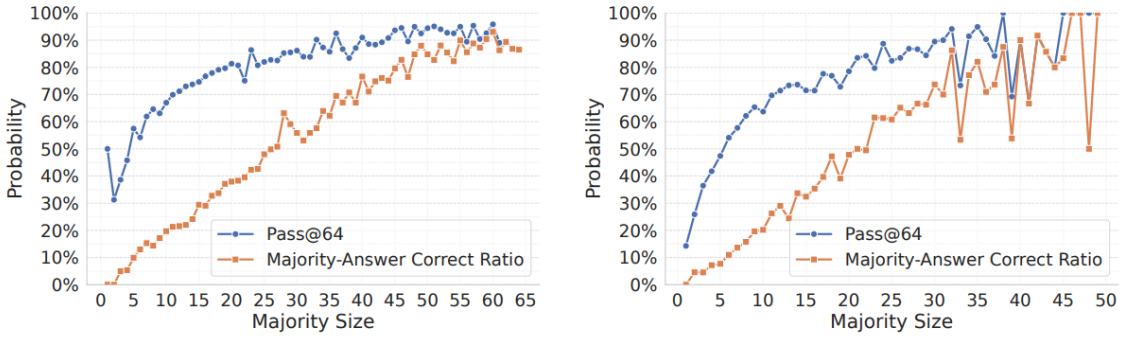
RESTRAIN: From Spurious Votes to Signals — Self-Driven RL with Self-Penalization

- Background: Unsupervised GRPO
  - Classically, the GRPO algorithm is supervised. That is to say, for each prompt x, there is a golden answer y^*. We collect $g$answers for $x$and calculate the reward by $\text{reward}(y_i, y^*)_{i=1}^{g}$.**However, like any labeling works, providing $y^*$is expensive and non-real-time.**
  - Thus, there is an unsupervised version of GRPO (TTRL). Here we don't rely on the golden answer. Instead, we use some mechanism to determine a fake label as the ground truth.
  - One common way to do this is major voting. That is, we, by default, regard the answer with highest frequency in the group as the fake label (ground truth).
  - However, as *the truth may belong to the minority*, this method may mislead the model. Thus, we have **RESTRAIN**, a framework to avoid the disadvantage.
- RESTRAIN
  - Pseudo-label weighting - *solve minority-correct by giving every answer a chance*



(a) Qwen3-4B-Base **(b)** Octothinker Hybrid 8B base

    - To achieve the goal of using *pass@n* mentioned in the caption above, we use *Pseudo-label weighting*.
    - Like the classic unsupervised GRPO, for each x, we let the model generate a group of answers of size n, $\{a_i\}_{i=0}^{n}$.
    - Then, we make a counter $\text{counter} = \{a_j : c_j\}_{j=1}^{m}$, where $m$is the # of unique answers, $c$is the number of occurrences of an answer a.
    - We regard **each** $a_j$as the ground truth, and respectively calculate the reward, advantage, and finally GRPO loss. We add the loss up with a weight $w$by:

$$L_{GRPO}(x;\theta) = \sum_{j=1}^{m} w_j \cdot L_{GRPO}(x, a_j; \theta)$$

    - The weight $w$is related to the occurrence frequency of each unique answer:

$$f_j = \frac{c_j}{n} \text{ is the frequency}$$

$$w_j = \frac{g(f_j)}{\sum_{l=1}^{m} g(f_l)}$$

where $g(\cdot)$ is a Gaussian function centered $\in [0, 1]$ with bias $\sigma > 0$.

  - Negative rollout penalization

- Pseudo-label weighting can solve the problem when *"the truth belongs to the minority"*. However, when the frequency of the "majority answer" is small (i.e., evenly distributed frequencies), **Pass@n drops significantly (meaning no correct answer exists in the pool)**, so Pseudo-label weighting will also be ineffective.
- Thus, a gate mechanism is designed. Let κbe the threshold of minimum majority frequency. Then:

$$\tilde{r}_{i,j} = \begin{cases} r_{i,j} & \text{if } M(x) \geq \kappa \\ 0 & \text{if } M(x) < \kappa \end{cases}, \quad \tilde{A}_{i,j} = \begin{cases} A_{i,j} & \text{if } M(x) \geq \kappa \\ A_{i,j} - \delta & \text{if } M(x) < \kappa \end{cases}$$

  This is to say, if $M(x)$(for prompt x, the frequency of the majority answer) does not reach κ, we regard all answers as "bad", so set the reward to 0. However, as the advantage, in this case, is 0, to update the model, we subtract a $\delta > 0$from the advantage to penalize all rollouts with low self-consistency.

- Prompt-level weighting
  - Some tasks are easy, while others are hard. We let a **frozen** reference model ϕ evaluate the difficulty of the questions **prior** to the RL process, and the difficulty value is constant during the RL.
  - Let ϕgenerate a set of answers of size n, and let $c_\phi$be the majority frequency. Then the difficulty of weight is $u_x = g(\frac{c_\phi}{n})$.
  - Then add the weight to the loss:

$$L_{RESTRAIN}(x;\theta) = u_x \sum_{j=1}^{m} w_j \cdot L_{GRPO}(x, a_j; \theta)$$

  - Intuition: if a question is easy, the reference model will have high confidence, so the $M(x;\phi)$will be large. With a clear train signal, it is good to let the model pay more attention to this question. Otherwise, if the frequency of each question is approximately the same, i.e. $M(x;\phi)$is small, then the signal is more likely to be noise. In this way, the weight is small.

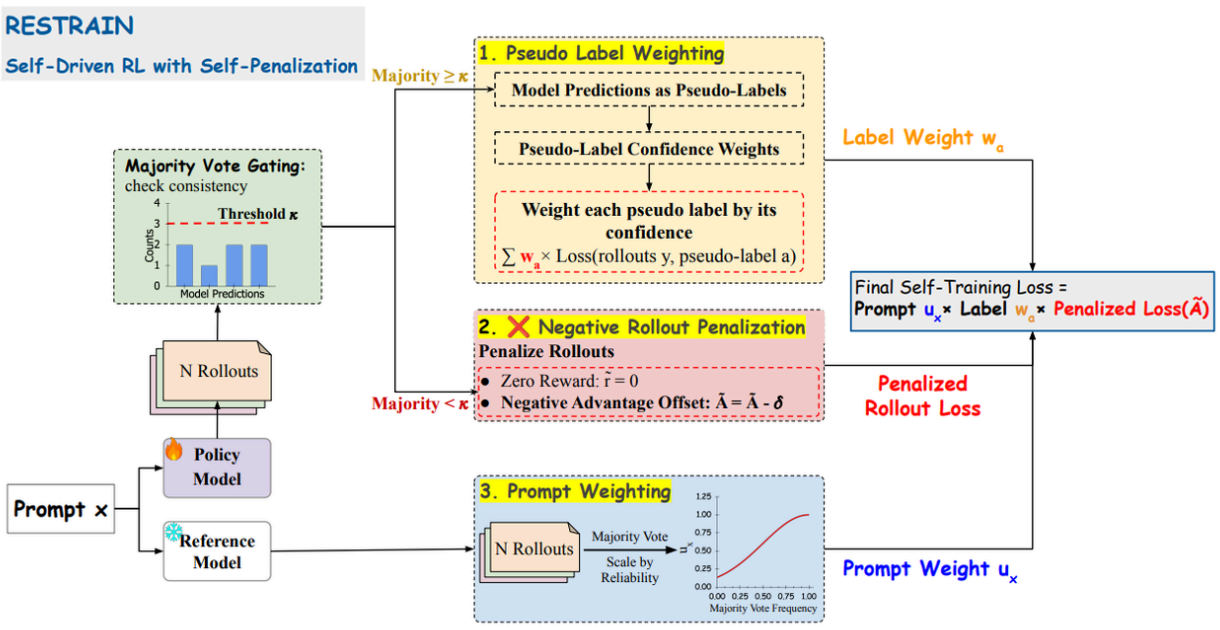


Figure 3 Overview of Our Method RESTRAIN: RESTRAIN consists of 3 core components: **1. Pseudo Label Weighting** which takes into account all possible model-predicted answers as candidate pseudo-labels when calculating final losses. **2. Negative Rollout Penalization** which penalizes rollouts with very low confidence by setting zero reward and applying negative advantage offsets to the losses. **3. Prompt Weighting** which downweights entire examples where the reference model predicts with low self-consistency.

- Experiment
    1. **Beats Unsupervised Baselines:** RESTRAIN significantly outperforms existing unsupervised methods (like TTRL and SRT). On the DAPO-14k dataset, it achieved **51.0%** accuracy compared to TTRL's 42.2%. It also showed consistent improvements on the fully synthetic S1k dataset.
    2. **Matches the "Gold Label" Upper Bound:** Remarkably, RESTRAIN nearly matches the performance of fully supervised training (using Gold Labels). The gap was only **0.4 points** (51.0% vs. 51.4%).
    3. **Better Generalization on Science Tasks:** It actually **surpassed** the Gold-Label supervised model on science-heavy benchmarks like **MMLU-STEM** (+7.2%) and **GPQA-Diamond** (+1.5%). This suggests that self-driven learning prevents overfitting to specific math solution patterns found in labeled data.
    4. **Prevents Model Collapse:** Standard unsupervised methods (like TTRL) tend to suffer from "mode collapse" (performance drops to zero) after about 50 training steps. RESTRAIN remains stable throughout training due to its soft weighting and penalization mechanisms.
    5. **Works for Test-Time Training:** The method is also effective for Test-Time Training (optimizing the model on the fly during inference), outperforming baselines like TTRL and ETMR by over 11%.
- Work Cited
    - Paper
    - Chat History